



PREMIUM

Full-Length Mock Test

Mock 02

Mathematics

GATE MA

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Abstract Algebra — Fields]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.2 [1 Mark] [EASY] [Real Analysis — Sequences]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.3 [1 Mark] [EASY] [Linear Algebra — Linear Maps]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.4 [1 Mark] [EASY] [Topology — Compactness]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.5 [1 Mark] [EASY] [ODE & PDE — First Order ODE]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.6 [1 Mark] [EASY] [Abstract Algebra — Homomorphisms]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.7 [1 Mark] [EASY] [Real Analysis — Series]

Kernel of group homomorphism is:

A) Subgroup

B) Normal subgroup

C) Quotient

D) Coset

Q.8 [1 Mark] [EASY] [Linear Algebra — Inner Products]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.9 [1 Mark] [EASY] [Topology — Compactness]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.10 [1 Mark] [EASY] [ODE & PDE — Wave Equation]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.11 [1 Mark] [EASY] [Abstract Algebra — Lagrange Theorem]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.12 [1 Mark] [EASY] [Real Analysis — Sequences]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.13 [2 Marks] [MODERATE] [Linear Algebra — Eigenvalues]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.14 [2 Marks] [MODERATE] [Topology — Connectedness]

A continuous function on closed bounded set is:

- A) Unbounded

B) Bounded

C) Nowhere

D) Discontinuous

Q.15 [2 Marks] [MODERATE] [ODE & PDE — Second Order ODE]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.16 [2 Marks] [MODERATE] [Abstract Algebra — Fields]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.17 [2 Marks] [MODERATE] [Real Analysis — Series]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.18 [2 Marks] [MODERATE] [Linear Algebra — Linear Maps]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.19 [2 Marks] [MODERATE] [Topology — Compactness]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.20 [2 Marks] [MODERATE] [ODE & PDE — PDEs]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.21 [2 Marks] [MODERATE] [Abstract Algebra — Fields]

Rank of identity matrix I_n :

- A) 0

B) 1

C) n

D) n^2

Q.22 [2 Marks] [MODERATE] [Real Analysis — Riemann Integral]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.23 [2 Marks] [MODERATE] [Linear Algebra — Diagonalization]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.24 [2 Marks] [MODERATE] [Topology — Compactness]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.25 [2 Marks] [MODERATE] [ODE & PDE — PDEs]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.26 [2 Marks] [MODERATE] [Abstract Algebra — Homomorphisms]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.27 [2 Marks] [MODERATE] [Real Analysis — Series]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.28 [2 Marks] [MODERATE] [Linear Algebra — Inner Products]

Kernel of group homomorphism is:

- A) Subgroup

B) Normal subgroup

C) Quotient

D) Coset

Q.29 [2 Marks] [MODERATE] [Topology — Continuity]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.30 [2 Marks] [MODERATE] [ODE & PDE — Wave Equation]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.31 [2 Marks] [MODERATE] [Abstract Algebra — Fields]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.32 [2 Marks] [MODERATE] [Real Analysis — Differentiability]

A continuous function on closed bounded set is:

- A) Unbounded
 - B) Bounded
 - C) Nowhere
 - D) Discontinuous
-

Q.33 [2 Marks] [HARD] [Linear Algebra — Linear Maps]

Rank of identity matrix I_n :

- A) 0
 - B) 1
 - C) n
 - D) n^2
-

Q.34 [2 Marks] [HARD] [Topology — Metric Spaces]

Kernel of group homomorphism is:

- A) Subgroup
 - B) Normal subgroup
 - C) Quotient
 - D) Coset
-

Q.35 [2 Marks] [HARD] [ODE & PDE — Laplace Transforms]

A continuous function on closed bounded set is:

- A) Unbounded

B) Bounded

C) Nowhere

D) Discontinuous

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Abstract Algebra — Fields]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.37 [2 Marks] [HARD] [Real Analysis — Series]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.38 [2 Marks] [HARD] [Linear Algebra — Inner Products]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.39 [2 Marks] [HARD] [Topology — Open/Closed Sets]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.40 [2 Marks] [HARD] [ODE & PDE — Laplace Transforms]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Abstract Algebra — Lagrange Theorem]**

Standard calculation using basic formula:

Answer: _____

Q.52 [2 Marks] [HARD] [Real Analysis — Continuity]

Standard calculation using basic formula:

Answer: _____

Q.53 [2 Marks] [HARD] [Linear Algebra — Vector Spaces]

Standard calculation using basic formula:

Answer: _____

Q.54 [2 Marks] [HARD] [Topology — Open/Closed Sets]

Standard calculation using basic formula:

Answer: _____

Q.55 [2 Marks] [HARD] [ODE & PDE — Laplace Transforms]

Standard calculation using basic formula:

Answer: _____

Q.56 [2 Marks] [HARD] [Abstract Algebra — Homomorphisms]

Standard calculation using basic formula:

Answer: _____

Q.57 [2 Marks] [HARD] [Real Analysis — Sequences]

Standard calculation using basic formula:

Answer: _____

Q.58 [2 Marks] [HARD] [Linear Algebra — Vector Spaces]

Standard calculation using basic formula:

Answer: _____

Q.59 [2 Marks] [HARD] [Topology — Continuity]

Standard calculation using basic formula:

Answer: _____

Q.60 [2 Marks] [HARD] [ODE & PDE — Wave Equation]

Standard calculation using basic formula:

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	B	Q.2:	B	Q.3:	C
Q.4:	B	Q.5:	B	Q.6:	C
Q.7:	B	Q.8:	B	Q.9:	C
Q.10:	B	Q.11:	B	Q.12:	C
Q.13:	B	Q.14:	B	Q.15:	C
Q.16:	B	Q.17:	B	Q.18:	C
Q.19:	B	Q.20:	B	Q.21:	C
Q.22:	B	Q.23:	B	Q.24:	C
Q.25:	B	Q.26:	B	Q.27:	C
Q.28:	B	Q.29:	B	Q.30:	C
Q.31:	B	Q.32:	B	Q.33:	C
Q.34:	B	Q.35:	B	Q.36:	AC
Q.37:	AC	Q.38:	AC	Q.39:	AC
Q.40:	AC	Q.41:	AC	Q.42:	AC
Q.43:	AC	Q.44:	AC	Q.45:	AC
Q.46:	AC	Q.47:	AC	Q.48:	AC
Q.49:	AC	Q.50:	AC	Q.51:	10
Q.52:	10	Q.53:	10	Q.54:	10
Q.55:	10	Q.56:	10	Q.57:	10
Q.58:	10	Q.59:	10	Q.60:	10

Detailed Solutions

Question 1 [Abstract Algebra — Fields]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 2 [Real Analysis — Sequences]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 3 [Linear Algebra — Linear Maps]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 4 [Topology — Compactness]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 5 [ODE & PDE — First Order ODE]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 6 [Abstract Algebra — Homomorphisms]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 7 [Real Analysis — Series]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 8 [Linear Algebra — Inner Products]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 9 [Topology — Compactness]

Rank of identity matrix I_n :

Answer:

Identity matrix I_n has rank n .

Question 10 [ODE & PDE — Wave Equation]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 11 [Abstract Algebra — Lagrange Theorem]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set S attains max/min.

Question 12 [Real Analysis — Sequences]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 13 [Linear Algebra — Eigenvalues]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 14 [Topology — Connectedness]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set S attains max/min.

Question 15 [ODE & PDE — Second Order ODE]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 16 [Abstract Algebra — Fields]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 17 [Real Analysis — Series]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set S attains max/min.

Question 18 [Linear Algebra — Linear Maps]

Rank of identity matrix I_n :

Answer:

Identity matrix I_n has rank n .

Question 19 [Topology — Compactness]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 20 [ODE & PDE — PDEs]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 21 [Abstract Algebra — Fields]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 22 [Real Analysis — Riemann Integral]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 23 [Linear Algebra — Diagonalization]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 24 [Topology — Compactness]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 25 [ODE & PDE — PDEs]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 26 [Abstract Algebra — Homomorphisms]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 27 [Real Analysis — Series]

Rank of identity matrix I_n :

Answer:

Identity matrix I_n has rank n .

Question 28 [Linear Algebra — Inner Products]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 29 [Topology — Continuity]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 30 [ODE & PDE — Wave Equation]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 31 [Abstract Algebra — Fields]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 32 [Real Analysis — Differentiability]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 33 [Linear Algebra — Linear Maps]

Rank of identity matrix I_n :

Answer: (C)

Identity matrix I_n has rank n .

Question 34 [Topology — Metric Spaces]

Kernel of group homomorphism is:

Answer: (B)

Kernel is always a normal subgroup.

Question 35 [ODE & PDE — Laplace Transforms]

A continuous function on closed bounded set is:

Answer: (B)

Extreme Value Theorem: continuous on compact set $!$ attains max/min.

Question 36 [Abstract Algebra — Fields]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 37 [Real Analysis — Series]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 [Linear Algebra — Inner Products]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 [Topology — Open/Closed Sets]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 [ODE & PDE — Laplace Transforms]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 [Abstract Algebra — Lagrange Theorem]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 [Real Analysis — Series]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 [Linear Algebra — Vector Spaces]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 [Topology — Open/Closed Sets]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 [ODE & PDE — PDEs]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 46 [Abstract Algebra — Rings]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 [Real Analysis — Continuity]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 [Linear Algebra — Vector Spaces]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 [Topology — Metric Spaces]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 [ODE & PDE — Laplace Transforms]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 [Abstract Algebra — Lagrange Theorem]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 [Real Analysis — Continuity]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 [Linear Algebra — Vector Spaces]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 [Topology — Open/Closed Sets]

Standard calculation using basic formula:

Answer:

Apply the appropriate formula for the given data.

Question 55 [ODE & PDE — Laplace Transforms]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56 [Abstract Algebra — Homomorphisms]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57 [Real Analysis — Sequences]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58 [Linear Algebra — Vector Spaces]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59 [Topology — Continuity]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60 [ODE & PDE — Wave Equation]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.
